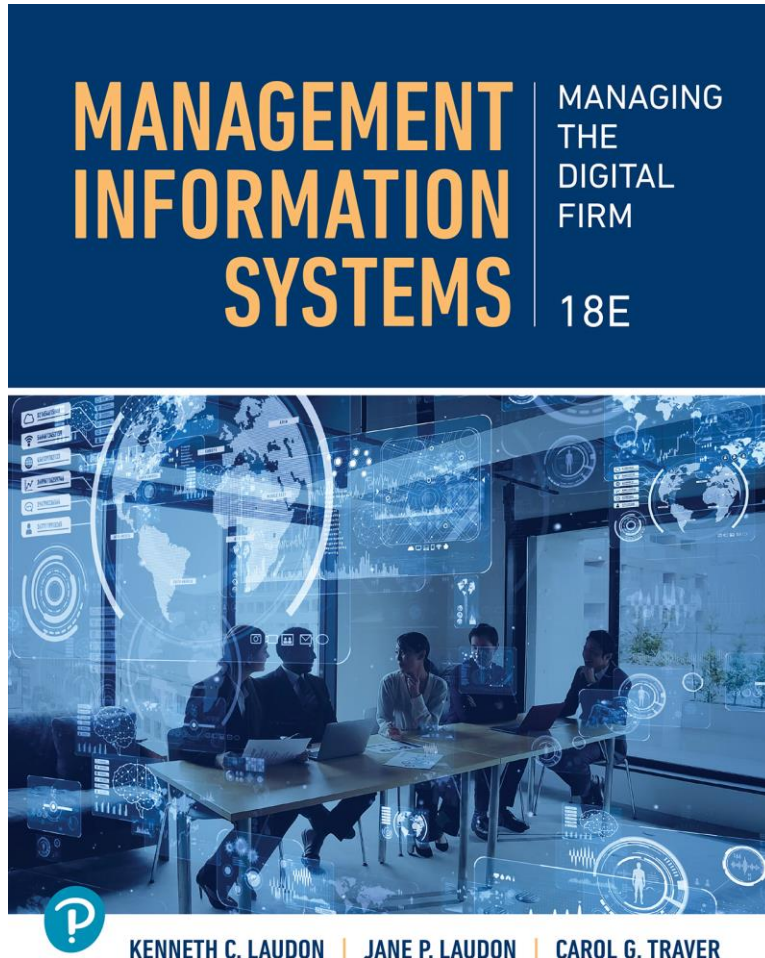


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 13

Building Information Systems

Learning Objectives

- 13.1** Explain how new information systems can promote organizational change.
- 13.2** Understand how IT can be used to manage and redesign business processes.
- 13.3** Describe the core activities in the systems development process.
- 13.4** Discuss methods for building and acquiring information systems.
- 13.5** Describe methods that enable rapid systems development.
- 13.6** Discuss the increasing democratization of software development.
- 13.7** Understand the impact of AI on software development.
- 13.8** Discuss approaches for building mobile applications.
- 13.9** Understand how the information in this chapter can help your career.

How New Information Systems Can Promote Organizational Change (1 of 3)

- Information technology can promote various degrees of organizational change, ranging from incremental to far-reaching
- The four kinds of structural organizational change that are enabled by information technology include
 - Automation
 - Rationalization of procedures
 - Business process redesign
 - Paradigm shifts

How New Information Systems Can Promote Organizational Change (2 of 3)

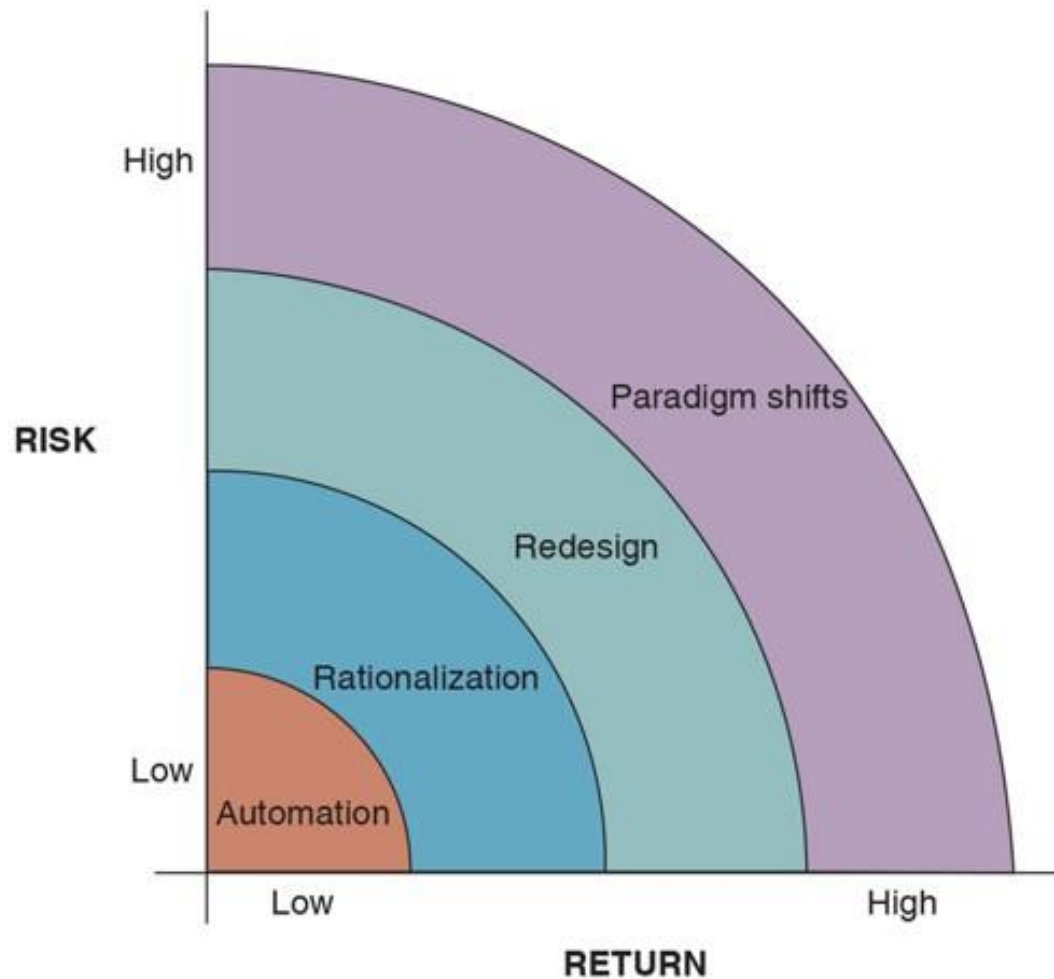
- Automation
 - Involves using IT to assist employees with performing tasks more efficiently and effectively
- Rationalization of procedures
 - Streamlining of standard operating procedures
 - Often found in programs for making a series of continuous quality improvements
 - Total quality management (TQM)
 - Six Sigma

How New Information Systems Can Promote Organizational Change (3 of 3)

- Business process redesign (BPR)
 - Analyze, simplify, and redesign business processes
 - Reorganize workflow, combine steps, eliminate repetition
- Paradigm shifts
 - Rethink nature of business
 - Define new business model
 - Change nature of organization

Figure 13.1

Organizational Change Carries Risks and Rewards



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How IT Can Be Used to Manage and Redesign Business Processes (1 of 2)

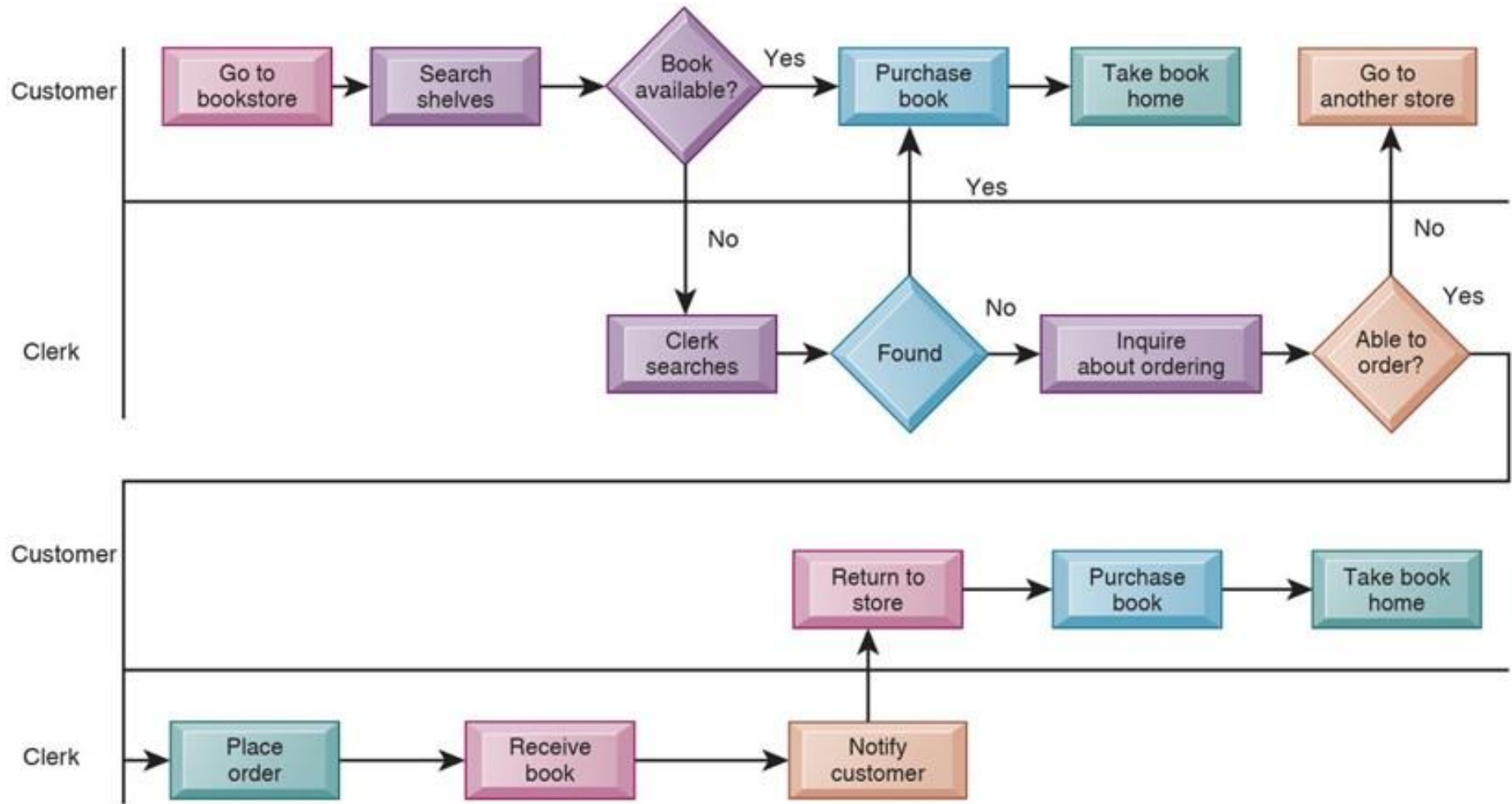
- Business process management (BPM)
 - Variety of tools and methodologies to analyze, design, and optimize those processes
 - BPM is never concluded because process improvement requires continual change

How IT Can Be Used to Manage and Redesign Business Processes (2 of 2)

- Companies practicing business process management go through the following steps
 1. Identify processes for change
 2. Analyze existing processes
 3. Design the new process
 4. Implement the new process
 5. Continuous measurement

Figure 13.2

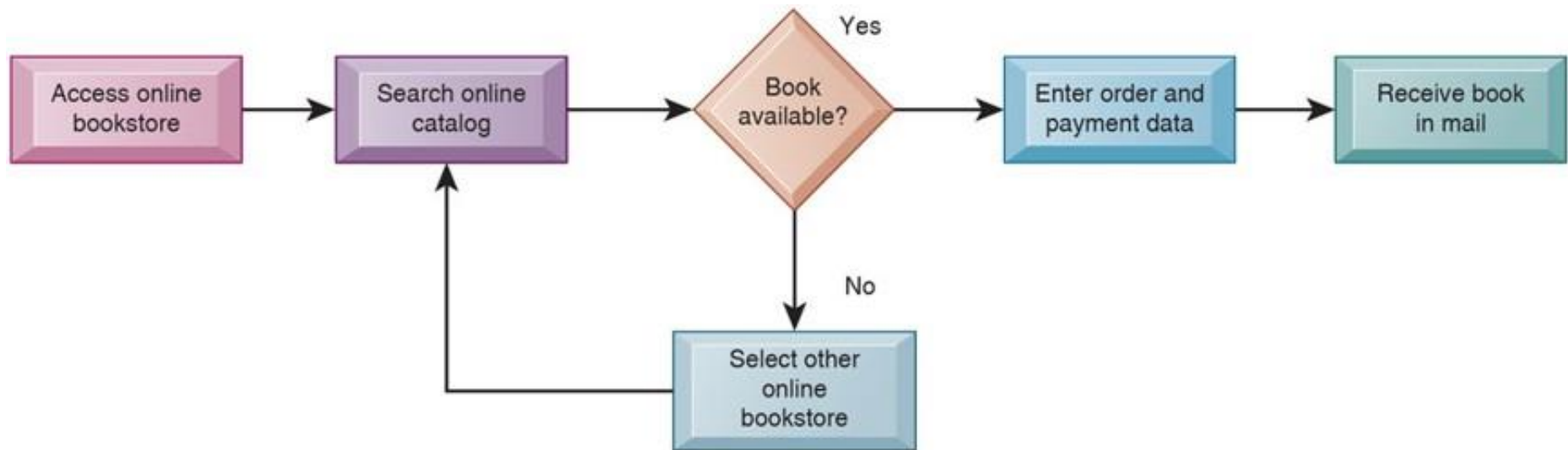
As-is Business Process for Purchasing a Book from a Physical Bookstore



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Figure 13.3

Redesigned Process for Purchasing a Book Online



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Tools for Business Process Management and Process Mining (1 of 2)

- There are many software tools for various aspects of BPM
 - IBM Workflow Automation, Oracle Business Process Management, and TIBCO BPM Enterprise
- These tools help businesses identify and document processes
 - Requiring improvement
 - Create models of improved processes
 - Capture and enforce business rules for performing processes
 - Integrate existing systems to support new or redesigned processes

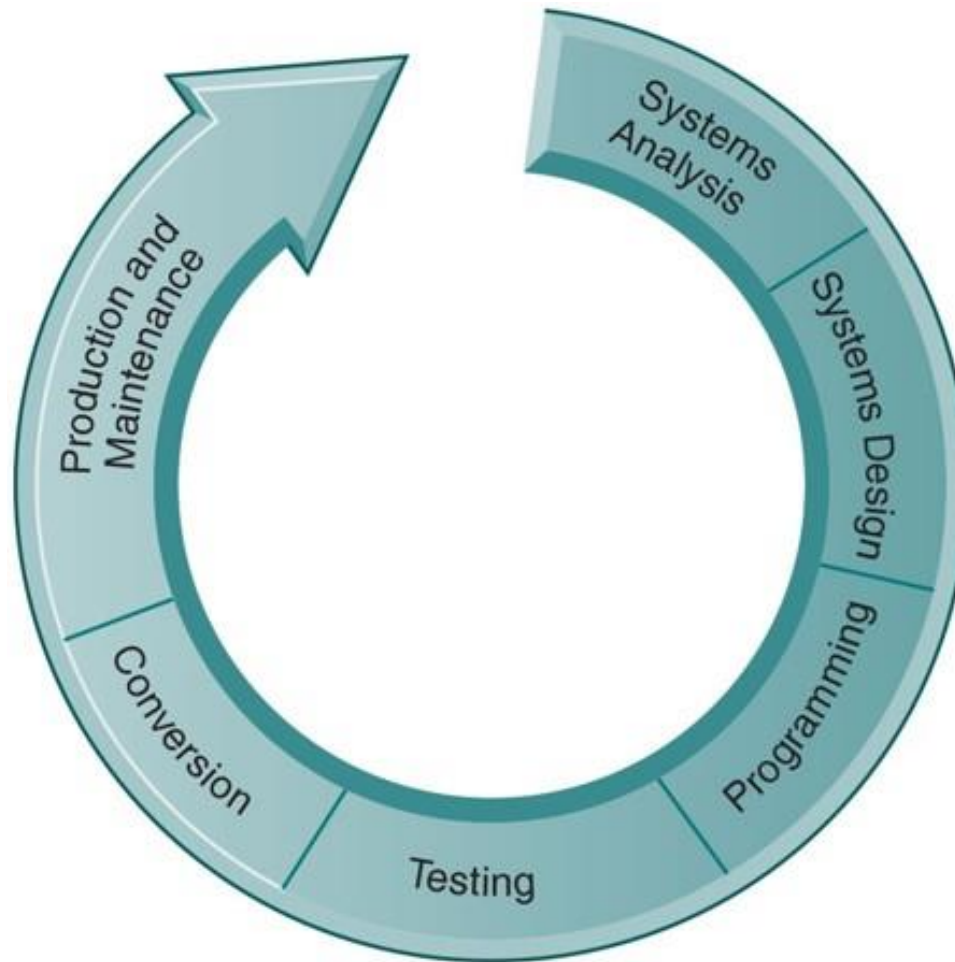
Tools for Business Process Management and Process Mining (2 of 2)

- Process mining
 - Analyzes data from a company's event logs to identify problems with certain processes and figure out ways to streamline them
- Enterprise information systems
 - Enterprise resource planning (ERP) software
 - Customer relationship management (CRM) software
 - Create audit trails of work processes that are recorded as log data

Core Activities in the Systems Development Process

- Systems development
 - Activities that go into producing an information system solution to an organizational problem or opportunity
 - Systems analysis
 - Systems design
 - Programming
 - Testing
 - Conversion
 - Production and maintenance

Figure 13.4 The Systems Development Process



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Systems Analysis

- Systems analysis
 - Analysis of a problem to be solved by a new system
 - Defining the problem
 - Identifying causes
 - Specifying solutions
 - Identifying information requirements
 - Feasibility study
 - Information requirements

Systems Design

- Systems design
 - Describes system specifications that will deliver functions identified during systems analysis
 - Should address all managerial, organizational, and technological components of system solution
 - Role of end users
 - User information requirements drive system building

Completing the Systems Development Process

(1 of 3)

- Programming
 - System specifications from design stage are translated into software program code
- Testing
 - Ensures system produces right results
 - **Unit testing**: Tests each program in system separately
 - **System testing**: Test functioning of system as a whole
 - **Acceptance testing**: Makes sure system is ready to be used in production setting
 - Test plan: All preparations for series of tests

Figure 13.5 A Sample Test Plan to Test a Record Change

Procedure		Address and Maintenance “Record Change Series”		Test Series 2		
Prepared by:			Date:		Version:	
Test Ref.	Condition Tested	Special Requirements	Expected Results	Output On	Next Screen	
2.0	Change records					
2.1	Change existing record	Key field	Not allowed			
2.2	Change nonexistent record	Other fields	“Invalid key” message			
2.3	Change deleted record	Deleted record must be available	“Deleted” message			
2.4	Make second record	Change 2.1 above	OK if valid	Transaction file	V45	
2.5	Insert record		OK if valid	Transaction file	V45	
2.6	Abort during change	Abort 2.5	No change	Transaction file	V45	

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Completing the Systems Development Process

(2 of 3)

- Conversion
 - Process of changing from an old system to a new system
 - **Four main strategies**
 - Parallel strategy
 - Direct cutover strategy
 - Pilot study strategy
 - Phased approach strategy
 - Requires end-user training and documentation
 - Finalization of detailed documentation showing how system works from technical and end-user standpoint

Completing the Systems Development Process

(3 of 3)

- Production and maintenance

- Production

- After conversion, the system is said to be in production
 - Post-implementation audit

- Maintenance

- Changes in hardware, software, documentation, or procedures to a production system to correct errors, meet new requirements, or improve processing efficiency

Methods for Building and Acquiring IS

(1 of 4)

- Systems differ in terms of their size and technological complexity
 - Also in terms of the organizational problems they are meant to solve
- A number of systems-building approaches have been developed to deal with these differences

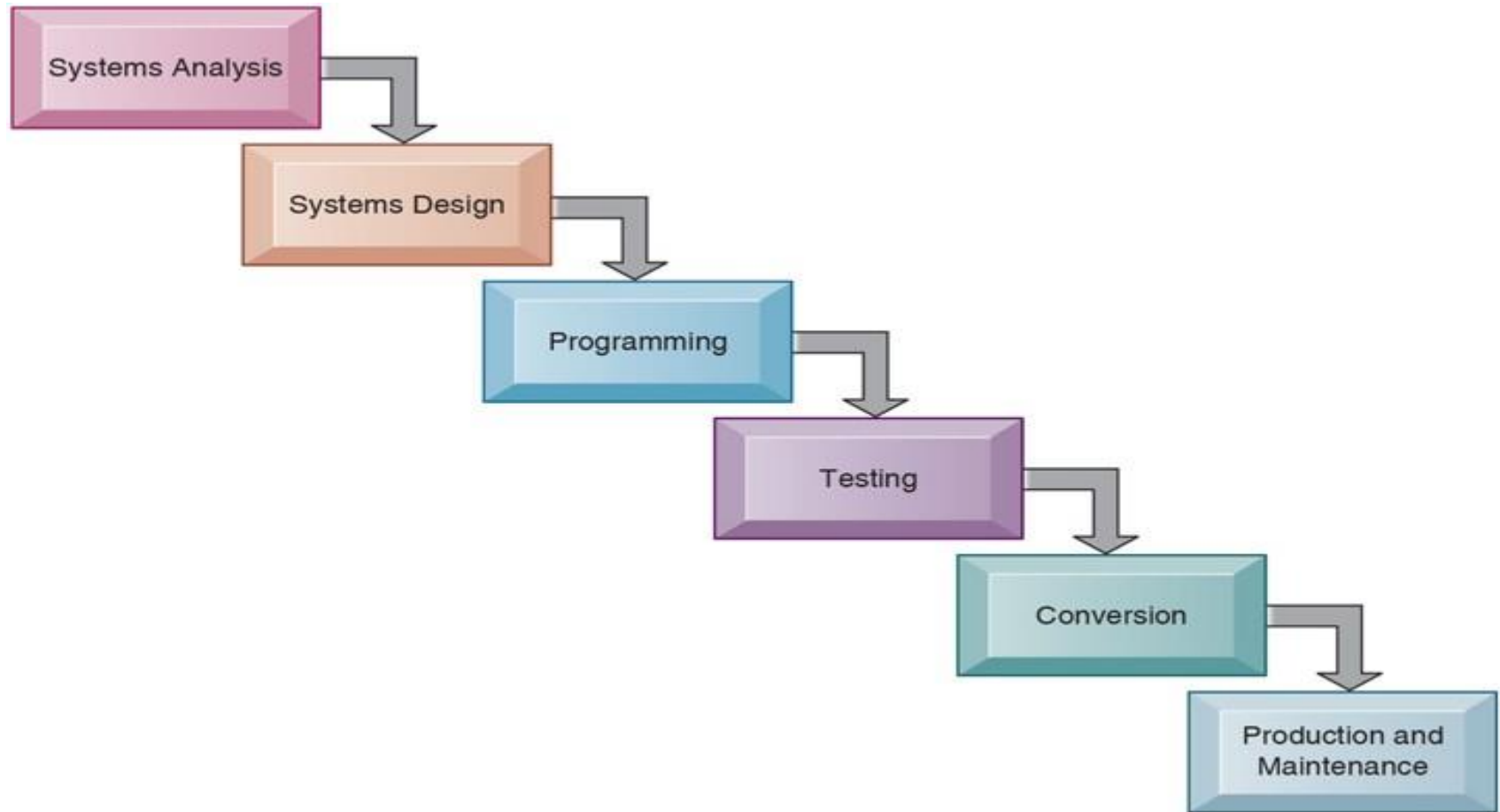
Methods for Building and Acquiring IS

(2 of 4)

- Systems development life cycle (SDLC)
 - Traditional method for building information systems
 - Uses a phased approach to building a system, dividing into a series of sequential formal stages
 - Systems analysis
 - Systems design
 - Programming
 - Testing
 - Conversion
 - Production and maintenance

Figure 13.6

The Traditional Systems Development Life Cycle



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Methods for Building and Acquiring IS

(3 of 4)

- Prototype

- A working version of an information system but meant to be a preliminary model

- Prototyping

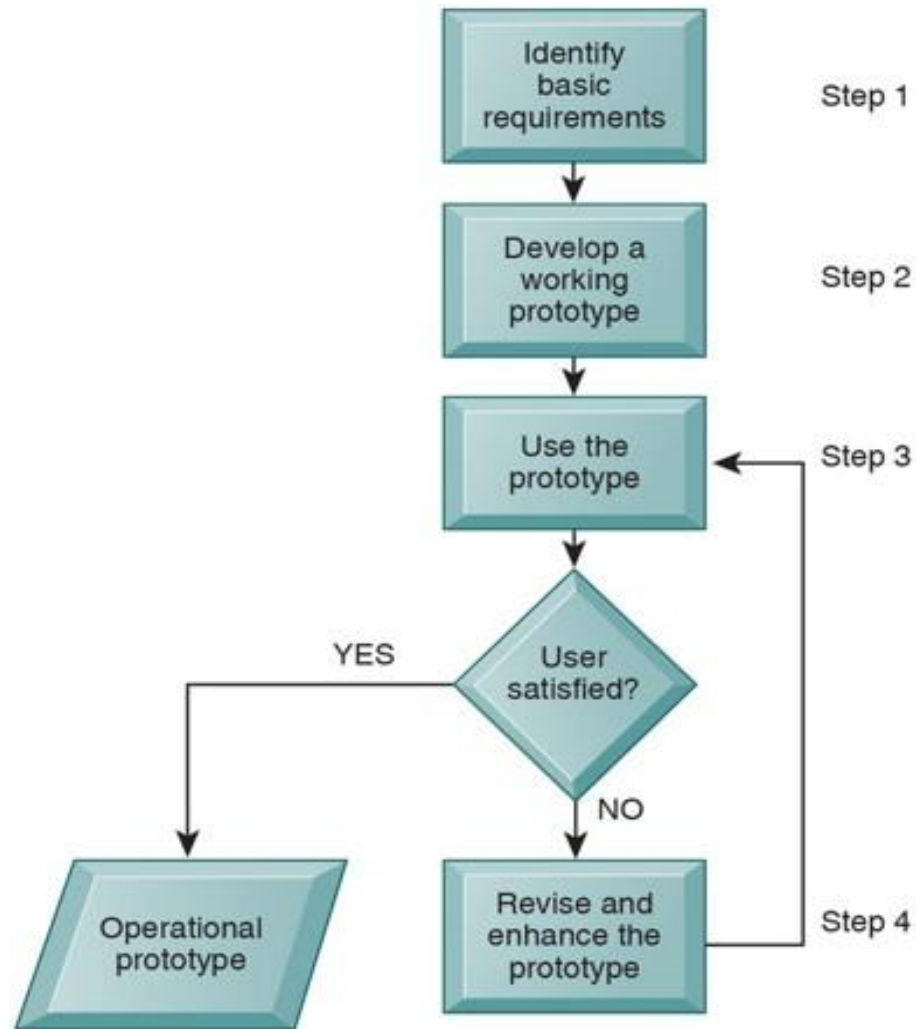
- Building experimental system rapidly and inexpensively for end users to evaluate
- Steps in prototyping:
 1. Identify the user's basic requirements
 2. Develop an initial prototype
 3. Use the prototype
 4. Revise and enhance the prototype

Methods for Building and Acquiring IS

(4 of 4)

- Advantages of prototyping
 - Useful if some uncertainty in requirements or design solutions
 - Often used for end-user interface design
 - More likely to fulfil the end-user requirements
- Disadvantages of prototyping
 - May gloss over essential steps
 - May not accommodate large quantities of data or large no. of users
 - May not undergo full testing or documentation

Figure 13.7 The Prototyping Process



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Application Software Packages, Software Services, and Outsourcing (1 of 3)

- Application software packages and software services
 - Save time and money
 - Many packages offer customization features
- Evaluation criteria for systems analysis include
 - Functions provided, flexibility, user friendliness, required resources, database requirements, installation and maintenance efforts, documentation, vendor quality, and cost
- Request for proposal (RFP)
 - Detailed list of questions submitted to software vendors
 - Used to evaluate alternative software packages and cloud software services

Application Software Packages, Software Services, and Outsourcing (2 of 3)

- Outsourcing

- Cloud and SaaS providers: subscribing companies use software and hardware provided by vendors
- External vendors
 - Hired to design, create software
 - Domestic outsourcing
 - Driven by firm's need for additional skills, resources, assets
 - Offshore outsourcing
 - Driven by cost-savings

Application Software Packages, Software Services, and Outsourcing (3 of 3)

- Outsourcing Advantages/Disadvantages
 - Advantages: Allows organization flexibility in IT needs
 - Disadvantages
 - Hidden costs, for example:
 - Identifying and selecting vendor
 - Transitioning to vendor
 - Opening up proprietary business processes to third party

Methods That Enable Rapid Systems Development

(1 of 3)

- In addition to using software packages and online software services, businesses are relying more heavily on fast-cycle techniques such as
 - Object-oriented design(OOD)
 - Rapid application development (RAD)
 - Joint application development (JAD)
 - Agile development, and
 - DevOps

Methods That Enable Rapid Systems Development

(2 of 3)

- **Object-oriented design (OOD)**

- Uses an object as the basic unit of systems analysis and design
- Object
 - Combines data and the specific processes that operate on the data

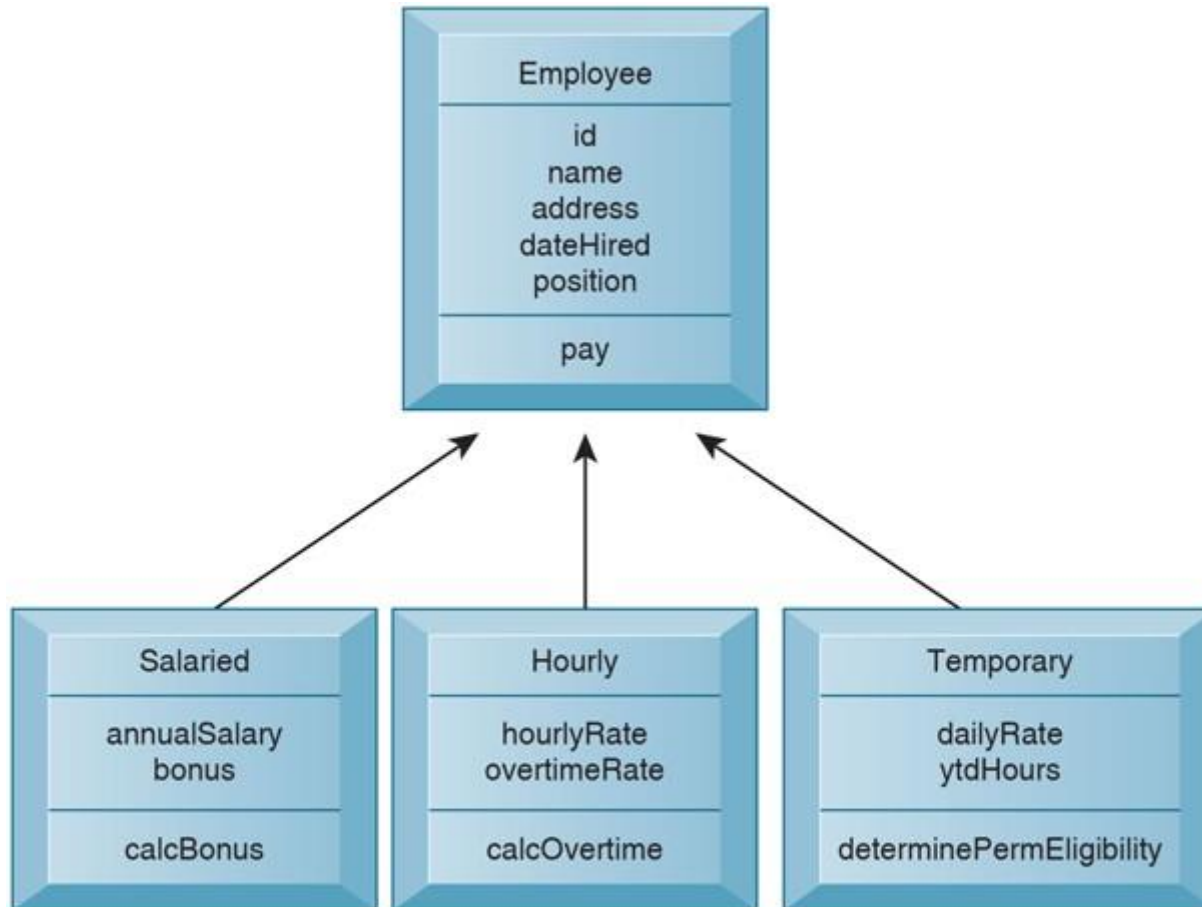
- **Rapid application development (RAD)**

- Process of creating workable systems in a very short period of time

- **Joint application design (JAD)**

- Used to accelerate generation of information requirements and to develop initial systems design

Figure 13.9 Class and Inheritance



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Methods That Enable Rapid Systems Development

(3 of 3)

- **Agile development**

- Focuses on rapid delivery of working software by breaking large project into subprojects that can be completed in short periods of time

- **DevOps**

- “Development and operations”
- Builds on agile development principles as an organizational strategy

Increasing Democratization of Software Development (1 of 3)

- One of the most significant trends impacting how information systems are built has been the increasing democratization of software development
 - Open-source software
 - Low-code software
 - No-code software

Increasing Democratization of Software Development (2 of 3)

- Open-source software (OSS)
 - Distributed with its source code, which is the part of the software that programmers manipulate to control how a program behaves
 - Programmers who have access to source code can modify a program
 - Powers systems and applications for tech giants like Google, Facebook, and Microsoft

Increasing Democratization of Software Development (3 of 3)

- Open-source software security
 - Although open-source software has many benefits, it introduces a whole new level of software risks
 - OSS is not completely bug-free

Low-Code and No-Code Development

- **Low-code development**

- Enables faster development of applications, with minimal hand-coding, using visual modelling in a graphical interface

- **No-code development**

- Includes built-in tools that enable user to create apps without any coding
- Some disadvantages
 - Have limited functionality
 - Users can create applications without proper management oversight

Citizen Development

- Citizen developer
 - A business user who builds new applications or modifies existing ones to create applications on their own with IT approval or with minimal help from IT staff

Impact of AI on Software Development

(1 of 2)

- AI has significantly impacted software development by
 - Enabling developers to automate routine tasks
 - Streamline workflows, and
 - Improve the accuracy and efficiency of software development processes

Impact of AI on Software Development

(2 of 2)

- The challenges of developing AI software include
 - The need for more software developer expertise in AI (currently in short supply)
 - The need for very large amounts of high-quality data to train AI models
 - AI as a Service (AlaaS) as part of the solution
 - Cloud-based machine learning platforms that offer scalable infrastructure and pre-built tools and frameworks for AI model development

CRISP-DM

- Standardize data mining processes across industries since 1999.
- The most common methodology for data mining, analytics, and data science projects.
- For best results, data science teams combine a loose implementation of CRISP-DM with overarching team-based agile project management approaches.



Reference: <https://www.datascience-pm.com/crisp-dm-2/>

Approaches for Building Mobile Applications

(1 of 3)

- Mobile website
 - A version of a regular website that is scaled down in content and navigation for easy access and search on a small screen
- Mobile web app
- Native app
 - A standalone application designed to operate using a specific mobile OS and device

Approaches for Building Mobile Applications

(2 of 3)

- Mobile web app
 - An Internet-enabled application with specific functionality for mobile devices
 - Designed to be accessed via the mobile device's browser
- Hybrid app
 - Many features of both a native app and a mobile webapp

Approaches for Building Mobile Applications

(3 of 3)

- Designing for a multiscreen world
 - Developing applications for mobile platforms is quite different from development for desktop computers and their much larger screens
 - Responsive web design (RWD)
 - Websites programmed so that layouts change automatically according to user's computing device
 - Adaptive web design (AWD)
 - A server hosting a website detects the attributes of the device making the request and loads a version of the site optimized for the device

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